Weekly Assignment: Relations (Epp Ch. 8)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 8.1 #4 c, d). Define a relation P on $\mathbb{Z}$ as follows: For every ordered pair $(m, n) \in \mathbb{Z} \times \mathbb{Z}$,

$$m \; P \; n \iff m \text{ and } n \text{ have a common prime factor.}$$

c. Is $0 \; P \; 5$?

d. Is $8 \; P \; 8$?

Problem 2 (Epp 8.1 #6 b, c). Let $X = \{a, b, c\}$. Define a relation S on $\mathscr{P}(X)$ as follows: For all sets A and B in $\mathscr{P}(X)$,

$$A \; S \; B \iff A \cap B \neq \varnothing.$$

b. Is $\{a, b\} \; S \; \{b, c\}$?

c. Is $\{a, b\} \; S \; \{a, b, c\}$?

Problem 3 (Epp 8.2 #10). C is the circle relation on the set of real numbers: For every $x, y \in \mathbb{R}$,

$$x \; C \; y \iff x^2 + y^2 = 1.$$

Determine whether C is reflexive, symmetric, transitive, or none of these. Justify your answers.

Problem 4 (Epp 8.2 #19). Define a relation I on $\mathbb{R}$ as follows: For all real numbers x and y,

$$x \; I \; y \iff x - y \text{ is irrational.}$$

Determine whether I is reflexive, symmetric, transitive, or none of these. Justify your answers.

Problem 5 (Epp 8.3 #14). Let A be the set of all strings of 0's, 1's, and 2's that have length 4 and for which the sum of the characters in the string is less than or equal to 2. A relation R is defined on A as follows: For every $s, t \in A$,

$$s \; R \; t \iff \text{the sum of the characters of } s \text{ equals the sum of the characters of } t.$$

Find the distinct equivalence classes of R.

Problem 6 (Epp 8.5 #6). Let P be the set of all people who have ever lived, and define a relation R on P as follows: For every $r, s \in P$,

$$r \; R \; s \iff r \text{ is an ancestor of } s, \text{ or } r = s.$$

Is R a partial order relation? Prove or give a counterexample.

Problem 7. Let R and S be relations on a set A. Prove or disprove: if R and S are transitive, then $R \cup S$ is transitive.